

Equations with brackets



- 1** Use substitution to determine whether the given value of x is the solution to each equation:
- a** $x = 6$ where $2(x - 3) = -8$. **b** $x = 8$ where $7(x - 6) = 14$.
c $x = 8$ where $3(x - 6) = 6$. **d** $x = 8$ where $9 = 7(x - 7)$.
- 2** Solve the following equations by first expanding the brackets:
- a** $2(x + 2) = 0$ **b** $7(x - 5) = 0$
c $4(x + 5) = 32$ **d** $5(x + 5) = 39$
e $4(x + 6) = -60$ **f** $12(x + 5) = 168$
g $3(x - 4) = -21$ **h** $4(x - 6) = -5$
i $5(x - 4) = -15$ **j** $12(x - 6) = 12$
k $5(2x + 6) = 69$ **l** $6(3x + 5) = 192$
m $9(3x + 5) = 153$ **n** $6(2x - 6) = -90$
o $6(3x - 5) = -102$ **p** $9(3x - 5) = 63$
q $-6(4x + 5) = -143$ **r** $-6(3x + 6) = 54$
s $-8(4x + 5) = 120$ **t** $-6(2x - 4) = 81$
u $-6(4x - 5) = 174$ **v** $-8(2x - 6) = -16$
- 3** Solve the following equations:
- a** $4(2x + 5) + 4 = 48$ **b** $6(4x + 8) - 9 = 87$
c $5(4x + 5) + 3x = 71$ **d** $4(2x - 6) - 3x = 16$
e $5(2x + 5) + 4x + 6 = 59$ **f** $4(4x - 6) - 3x + 8 = 36$
g $-3(4x + 7) + 3x - 5 = -8$ **h** $-4(4x + 7) - 3x + 5 = 15$
i $-3(4x + 6) - 3x - 7 = -70$ **j** $-4(4x + 6) + 3x + 7 = -56$
- 4** Solve the following equations:
- a** $4(3x + 5) + 3(2x + 6) = 74$ **b** $2(2x - 5) + 3(4x + 6) = 56$
c $2(4x + 5) + 3(3x - 6) = -42$ **d** $3(4x + 6) - 2(3x + 5) = 26$
e $5(4x - 6) - 3(2x + 5) = -73$ **f** $8(4x + 5) - 3(2x - 6) = -46$
g $-3(2x + 5) + 5(4x + 6) = 57$ **h** $-3(2x + 5) + 5(3x + 6) = 42$
i $-3(2x + 6) + 5(4x - 5) = -15$ **j** $-3(2x - 4) + 5(3x + 5) = 55$
k $-3(2x - 6) + 5(4x - 5) = 21$ **l** $-3(2x + 4) + 5(3x + 5) = 40$

Applications



- 5 A square has a side length of $4x + 5$ cm. If the perimeter of the square is 44 cm, find the value of x .

- 6 A rectangle has a width of $5x - 3$ cm and a height of $3x + 7$ cm. If the perimeter of the rectangle is 42 cm, find the value of x .